

EE 313: Measurement and Instrumentation

Course Code	EE 313
Course Name	Measurement and Instrumentation
L-T-P-C	2-1-0-3
Prerequisites	IC 160: Basic Electrical Engineering and IC 161 Basic Electronics
Intended for	UG
Distribution	Elective
Semester	5th and 7th semester
Approval	9th Senate (May 2016)

Course Contents

- **Fundamentals of measuring instruments:** Terminology, Units and standards, functional elements of an instrument, input-output configuration, static characteristics and static calibration - least squares calibration curves -static sensitivity - linearity, dynamic characteristics - dynamic response analysis - operational and sinusoidal transfer function - zero-, first-, and second-order instruments, Errors and handling of errors in measurement. [10 lectures]
- **Motion and dimensional measurement:** Relative displacement and velocity: translational and rotational - resistive potentiometers - resistance strain gauge - differential transformers - eddy current noncontacting transducers - ultrasonic transducers - mechanical flyball angular velocity sensor - mechanical revolution counters and timers, relative acceleration, accelerometers - deflection type - null balance type, gyroscopic angular-displacement and velocity sensors, laser rangefinders, ultrasonic rangefinders, radars for automobiles. [12 lectures]
- **Force, torque, and shaft power measurement:** Elastic force transducers - bonded-strain-gauge transducers - differential-transformer transducers - piezoelectric transducers - variable-reluctance, torque measurement on rotating shafts, dynamometers, vibrating-wire force transducers. [3 lectures]
- **Sound and Vibration measurement:** Deadweight gauges and manometers, elastic transducers, vibrating cylinder transducers, high- pressure measurement, low pressure measurement - diaphragm gauges - McLeod gauge - Knudsen gauge - momentum transfer gauges - thermal conductivity gauges - ionisation gauges - dual gauge technique, sound measurement - microphones - pressure response - acoustic intensity - acoustic emission. [3 lectures]
- **Temperature and heat-flux measurement:** Thermal expansion methods - bimetallic thermometers - liquid-in-glass thermometers - pressure thermometers, thermoelectric sensors, electrical resistance sensors - conductive - bulk semiconductor, junction semiconductor sensors, digital thermometers, radiation methods - detectors - automatic null balance - optical - two color - fluoroptic - infrared imaging. [5 lectures]
- **Manipulation, transmission and recording of data:** Bridge circuits, amplifiers, filters, integration and differentiation, cable transmission, fiber- optic data, radio telemetry, Pneumatic transmission, instrument connectivity, potentiometers, digital voltmeters and multimeters, electromechanical servotype XT and XY recorders, data acquisition systems. [4 lectures]
- **Electrical and Electronic Measurements:** Signal generation, Principle of LCR meter, Probe compensation, Spectrum analysis, Instrument Transformers, Measurement of Power and Wattmeters, Measurement of Energy and Industrial Metering, Digital Storage Oscilloscope, Issues of sampling, memory, ADC speed. [6 lectures]

Text Books

1. E. O. Doebelin, D. N. Manik, **Measurement systems Application and Design**, 5th Edition, McGraw Hill Book Company, 2007.
2. D. Patranabis, **Principles of Industrial Instrumentation**, 3rd Edition, Tata McGraw Hill Publishing Ltd., 2010.

References

1. A.K. Sawhney, **A course in electrical and electronic measurements and instrumentation**, 19th Edition, Dhanpat Rai Publications, 2011.
2. A. S. Morris, **Measurement and Instrumentation Principles**, 3rd Edition, Butterworth-Heinemann, 2001.
3. J. P. Holman, **Experimental Methods for Engineers**, McGraw Hill Book Company, 1971.
4. W. C. Dunn, **Fundamentals of Industrial Instrumentation and Process Control**, McGraw Hill, 2005.